

Commissioning of Phase 2 and Phase 3 SuperKEKB

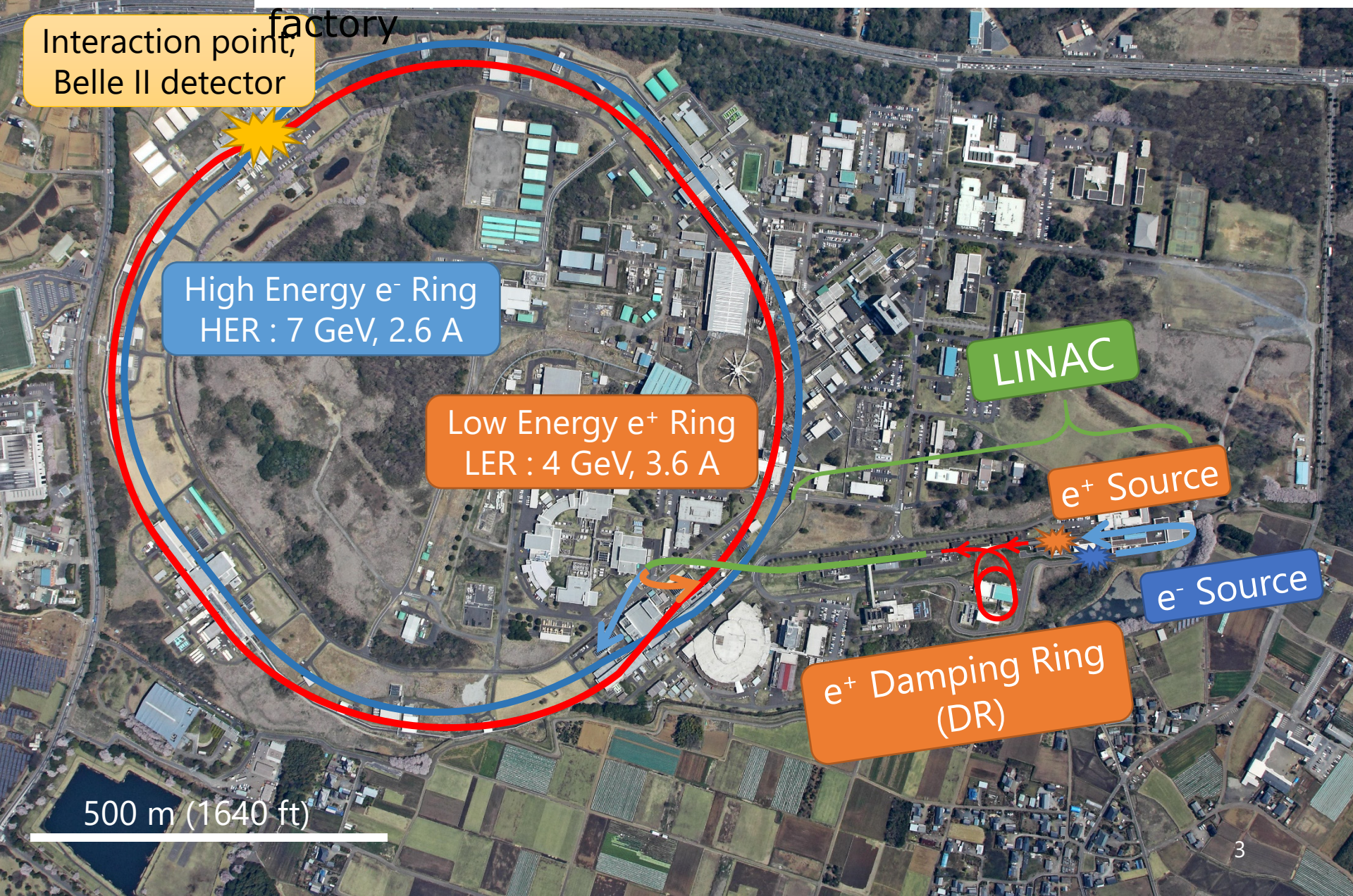
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for the SuperKEKB Commissioning Group

9/2/2019
NAPAC2019

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SuperKEKB: asymmetric-energy electron-positron collider constructed by upgrading the KEKB B-



Machine Design Parameters

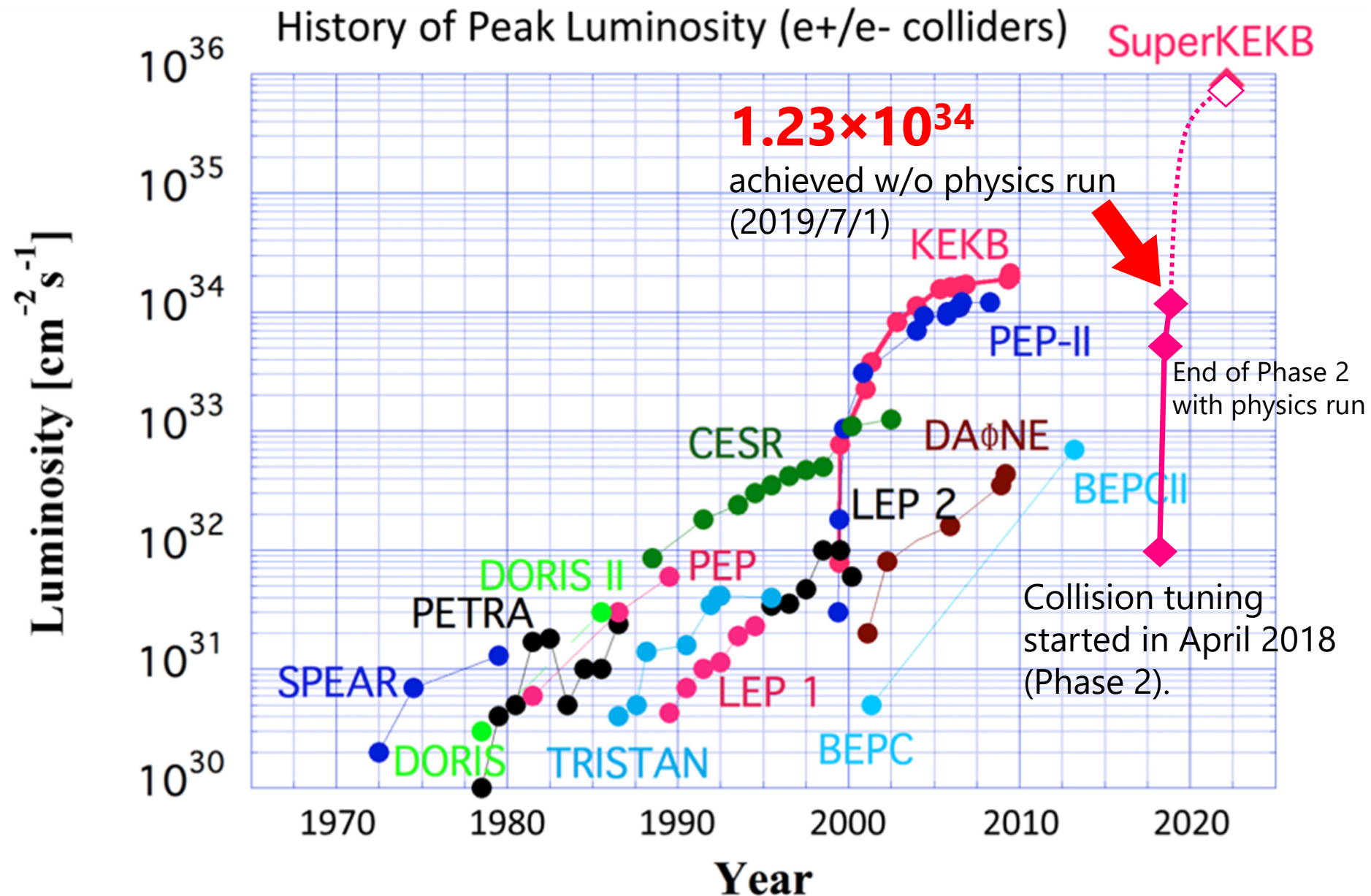
2017/September/1	LER	HER	unit	
E	4.000	7.007	GeV	
I	3.6	2.6	A	
Number of bunches	2,500			
Bunch Current	1.44	1.04	mA	
Circumference	3,016.315		m	
ϵ_x/ϵ_y	3.2(1.9)/8.64(2.8)	4.6(4.4)/12.9(1.5)	nm/pm	() : zero current
Coupling	0.27	0.28		includes beam-beam
β_x^*/β_y^*	32/0.27	25/0.30	mm	
Crossing angle	83		mrاد	
α_p	3.20×10^{-4}	4.55×10^{-4}		
σ_δ	$7.92(7.53) \times 10^{-4}$	$6.37(6.30) \times 10^{-4}$		
V_c	9.4	15.0	MV	
σ_z	6(4.7)	5(4.9)	mm	
v_s	-0.0245	-0.0280		
v_x/v_y	44.53/46.57	45.53/43.57		
U_0	1.76	2.43	MeV	
$\tau_{x,y}/T_s$	45.7/22.8	58.0/29.0	msec	
ξ_x/ξ_y	0.0028/0.0881	0.0012/0.0807		
Luminosity	8×10^{35}		$\text{cm}^{-2}\text{s}^{-1}$	

Beam current is
Two times larger than KEKB
×

Beam size at collision point is
20 times smaller than KEKB

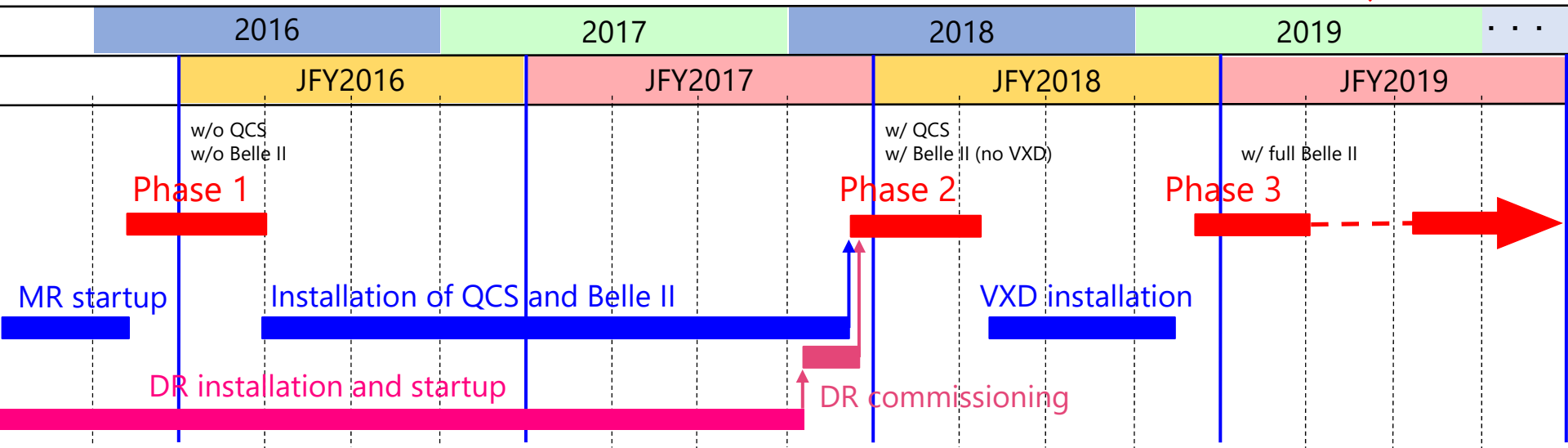


Target Luminosity is
 $8 \times 10^{35} \text{ cm}^{-2}\text{s}^{-1}$,
40 times higher than KEKB.



Commissioning in a phased manner

We are here



Phase 1 :

Basic machine test without collision /
Low emittance beam tuning / Vacuum scrubbing

Phase 2 :

Squeezing beta at IP / "nano-beam scheme" collision tuning /
Physics run w/o VXD starts with BG study

Phase 3 :

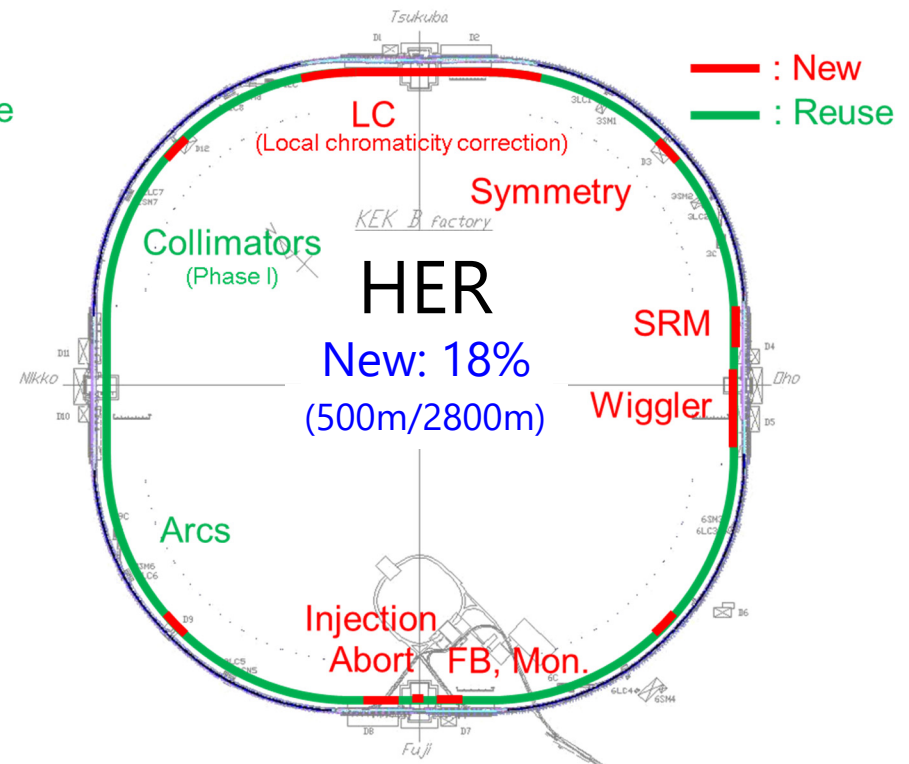
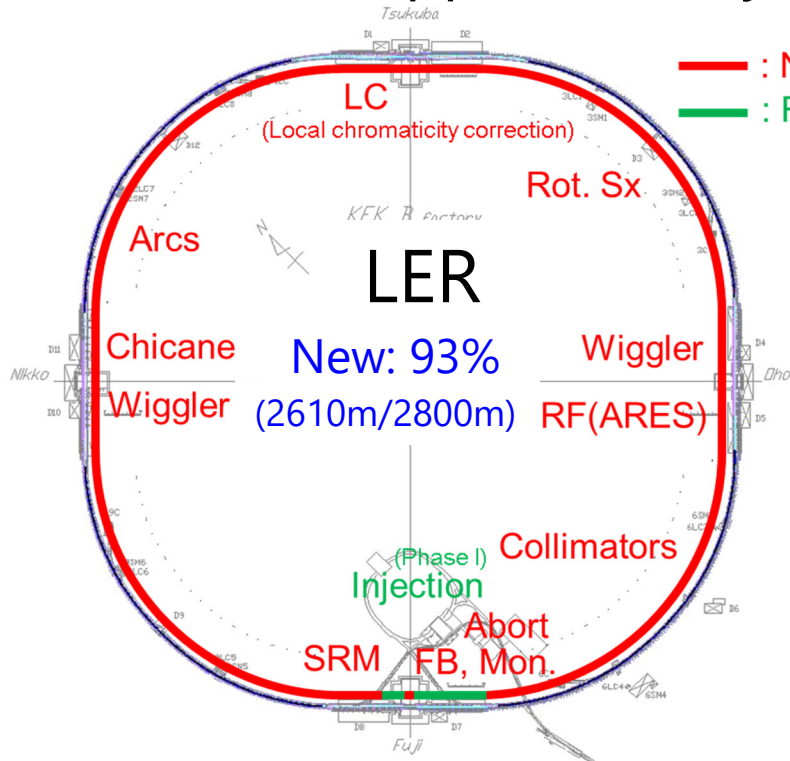
Physics run starts with a fully equipped Belle II detector
Further collision tuning and collimator tuning / cont. inj.

Phase 1 commissioning result

- Completion of the SuperKEKB MR without the beam interaction region and the injector LINAC without the DR.
- Vacuum scrubbing was done and beam background studies were performed by BEAST detector.
- Optics correction was performed for achieving low vertical emittance.
 - ✓ The HER corrections went well.
 - ✓ In the LER corrections, the leakage magnetic field from the Lambertson septum magnets for beam abort turned out to have a harmful effect on the emittance.
 - ▣ The problem was resolved by the skew quadrupole corrector and the permanent skew quadrupole magnet.
- Vertical beam size blowup was observed in LER. The cause was the electron clouds.
- Injector LINAC work stably.

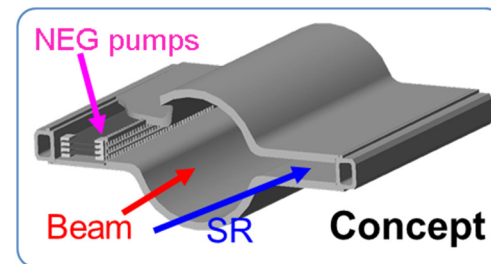
Vacuum scrubbing

- LER: Approximately 93% of beam pipes in length are new.
- HER: Approximately 82 % are reused.



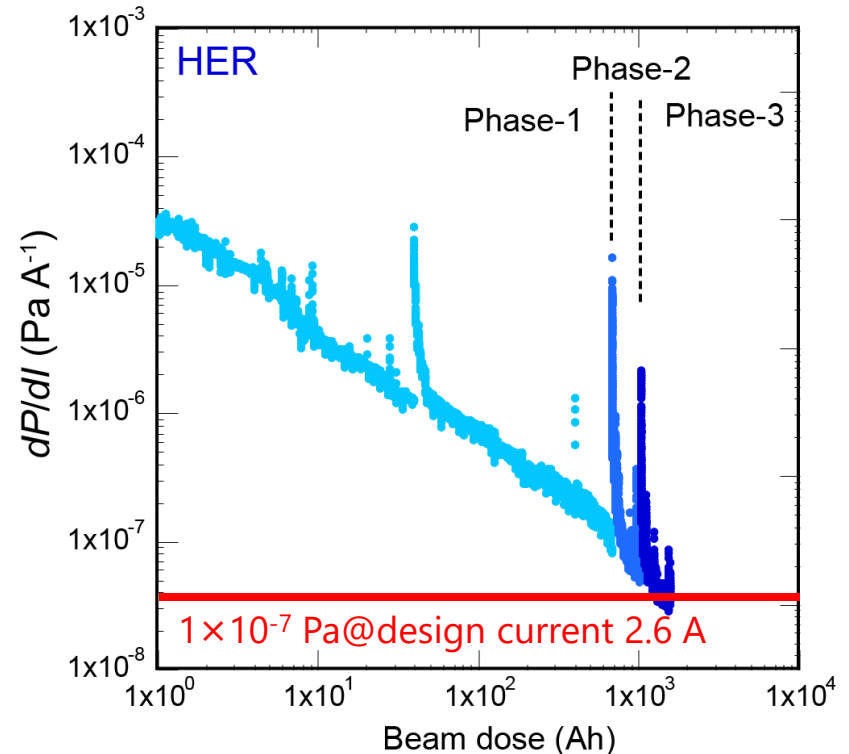
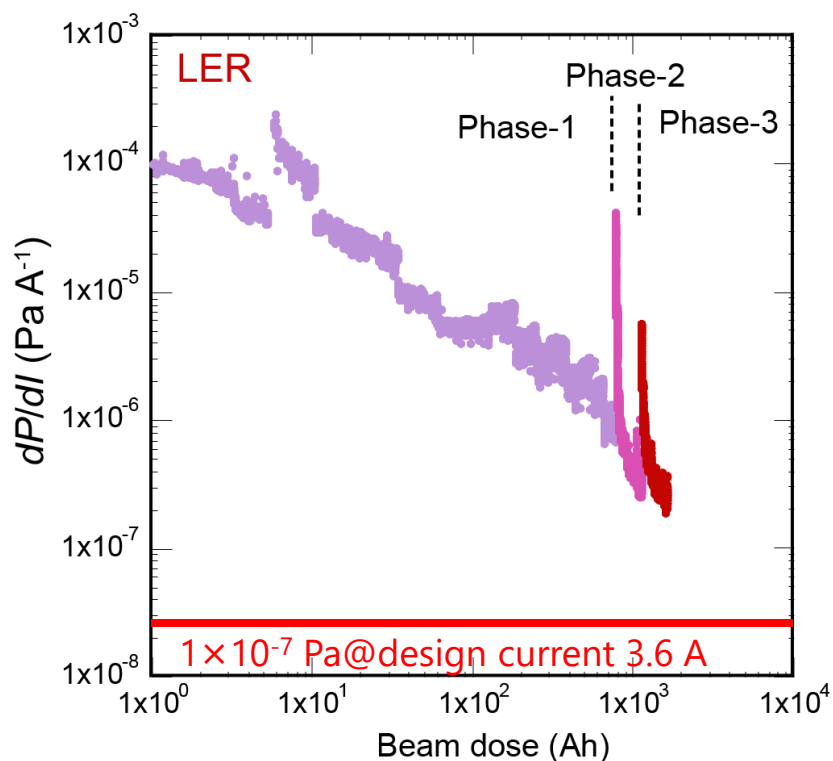
Typical size: $\phi 90 \times w220$

- New beam pipes are designed for suppression of electron cloud effect to avoid beam size blowup. (antechamber, TiN coating, etc.)



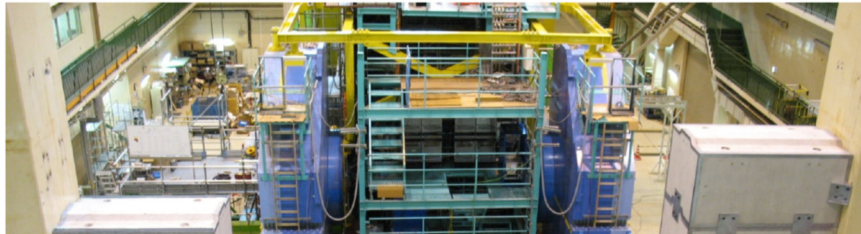
Status of ring vacuum pressures

Average ring vacuum pressures normalized by a unit beam currents.



- dP/dI should be kept lower than **red lines**, in order to increase beam lifetime and decrease beam-gas detector background.
- HER vacuum pressure has reached the target value.
- Further scrubbing (5 kAh for 1/3 pressure) is necessary for LER, where newly installed beam pipes are used.

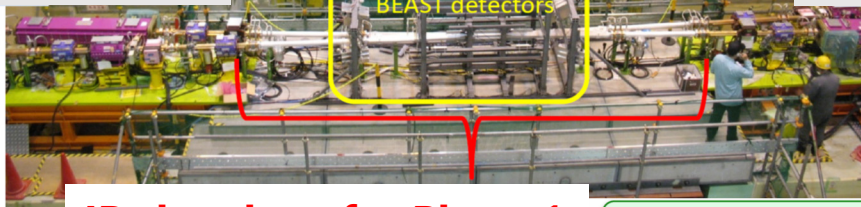
Construction of Interaction Region between Phase 1 and 2



Installation of beam final focus superconducting magnet system (QCS; QCSL, QCSR) into IR.

R-side
(FWD, D2-side, Oho-side)

L-side
(BWD, D1-side, Nikko-side)

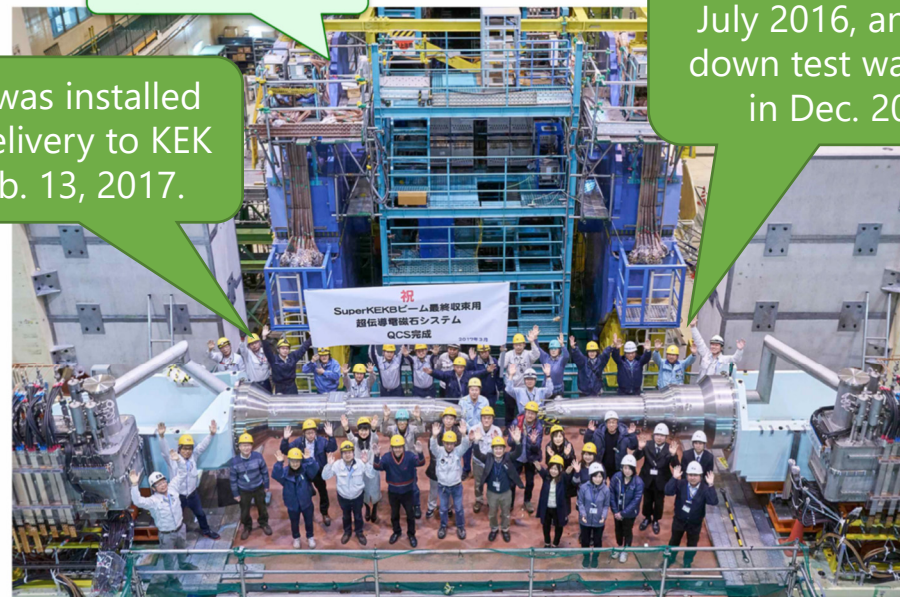


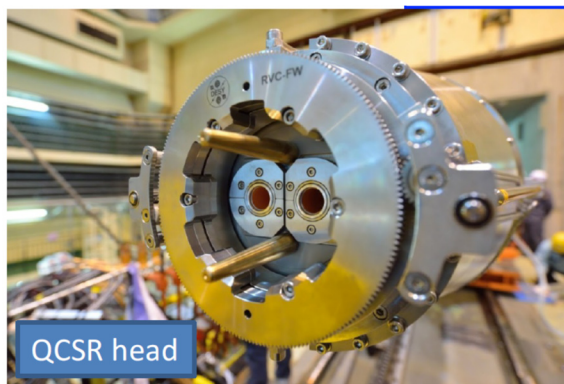
IP chambers for Phase 1

Belle II detector before Roll-in.

QCSR was installed after delivery to KEK on Feb. 13, 2017.

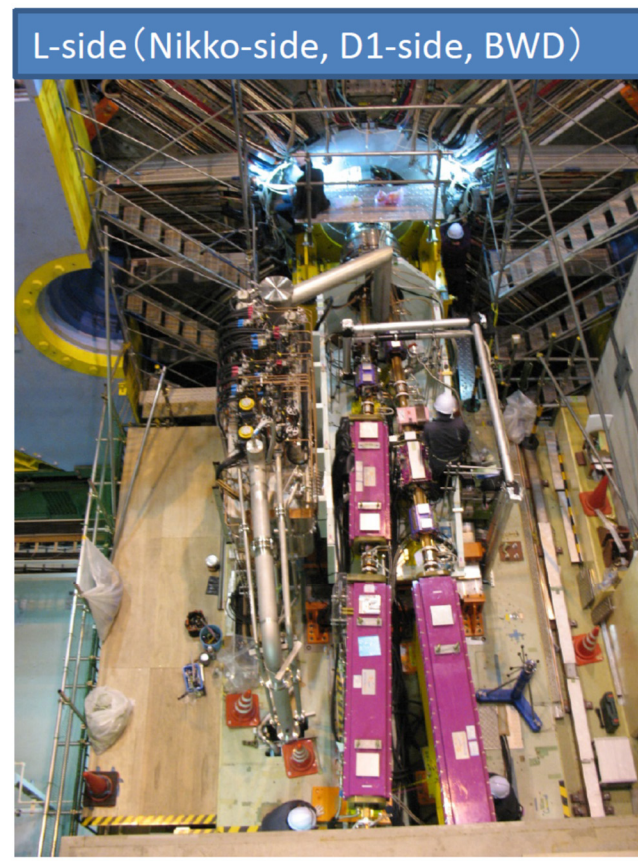
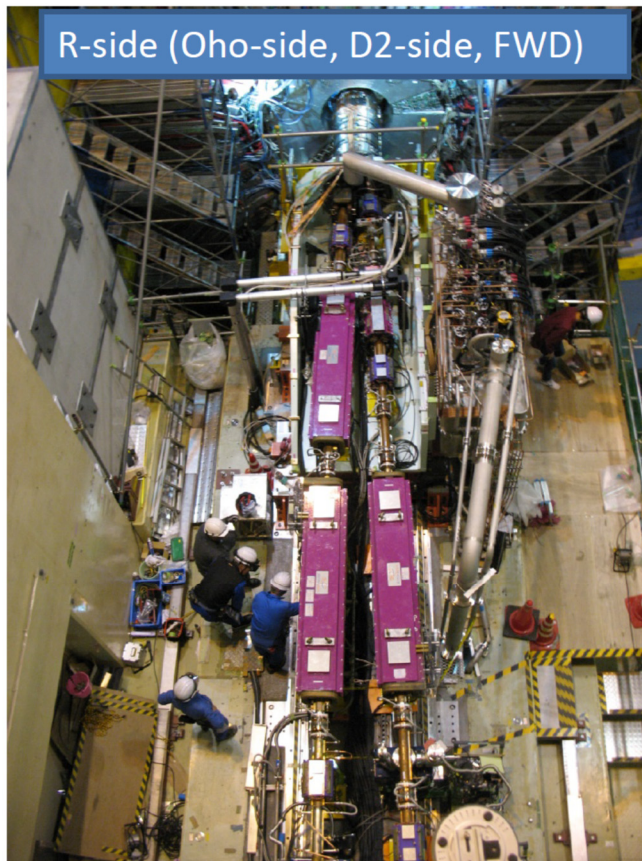
QCSL was installed in July 2016, and cool down test was done in Dec. 2016.





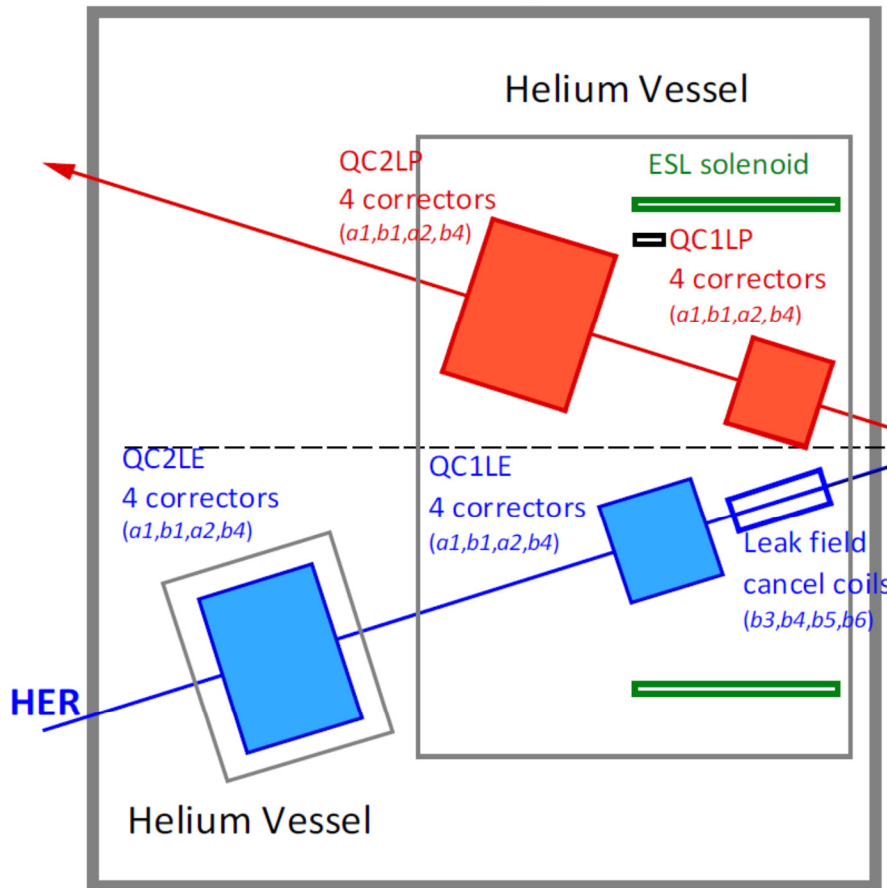
Connection between the IP chamber and QCS beam pipes by remote operation. The connection on Jan. 15, 2018 worked well.

All magnets were re-installed and all beam pipes were connected. Cabling and other tasks finished in March, 2018.

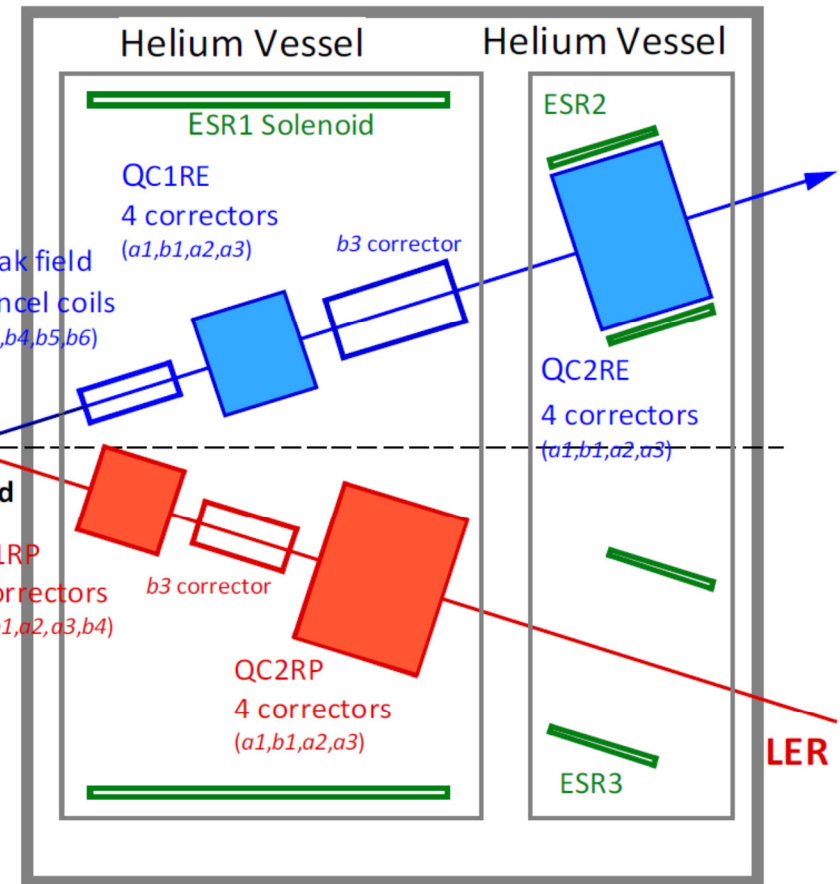


Final focus superconducting magnet (QCS)

QCS-L Cryostat



QCS-R Cryostat



- 4 compensation solenoids
- 8 main quadrupole SC magnets
- 35 corrector SC magnets
- 8 leak field cancel SC magnets

Power supplies for QCS



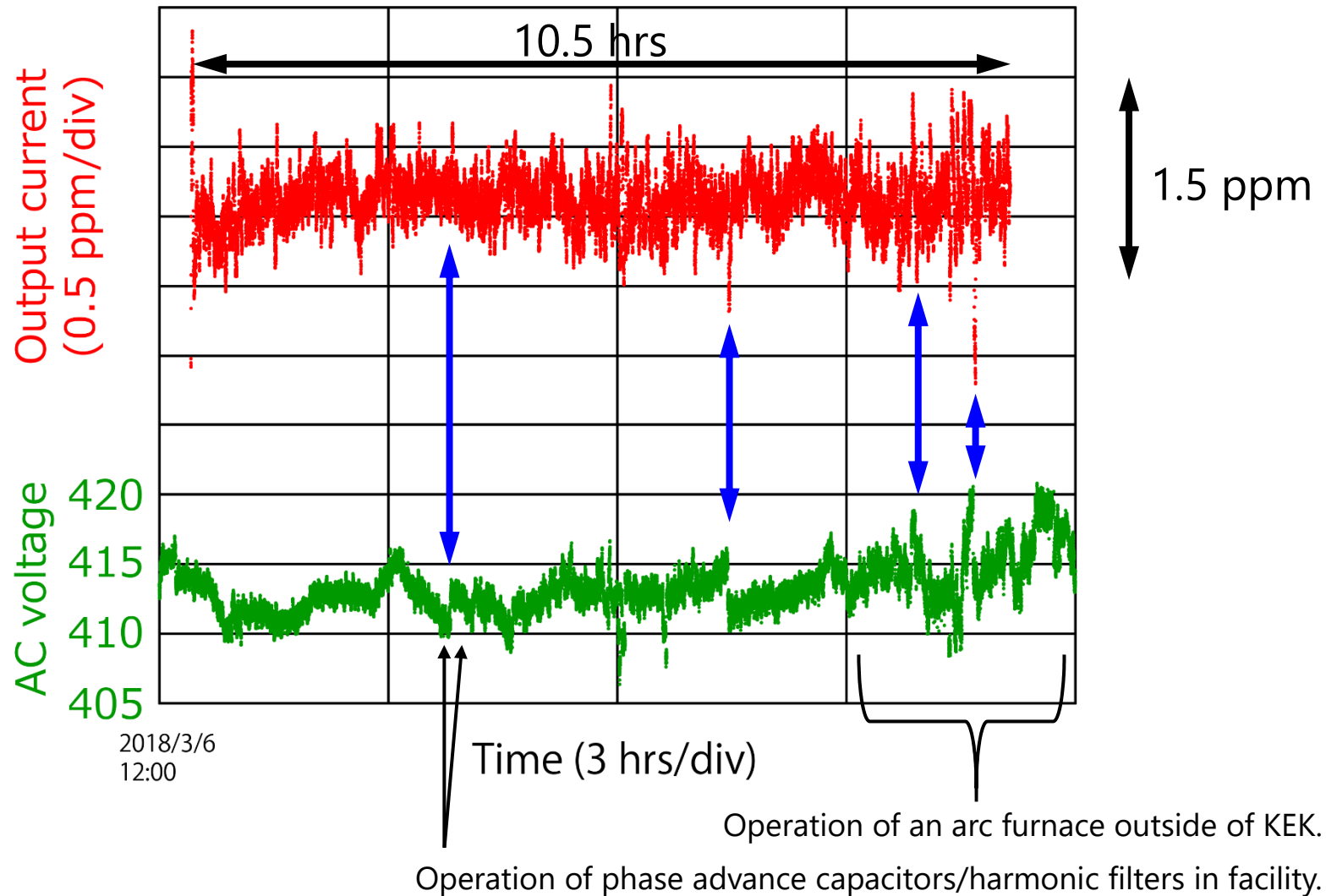
Main quadrupole magnet:
Stability : 2 ppm/8 hours
Ripple: 1 ppm

Correction coils:
Stability : 5 ppm/8 hours

Magnet	Rated Output	# of PS's
Main Quads	2000A, 15V	8
Correctors	$\pm 70\text{A}$, $\pm 10\text{V}$	43
ESL	410A, 30V	1
ESR1	455A, 45V	1
ESR2 & 3	155A, 15V	1



Performance test results: QC2RE stability

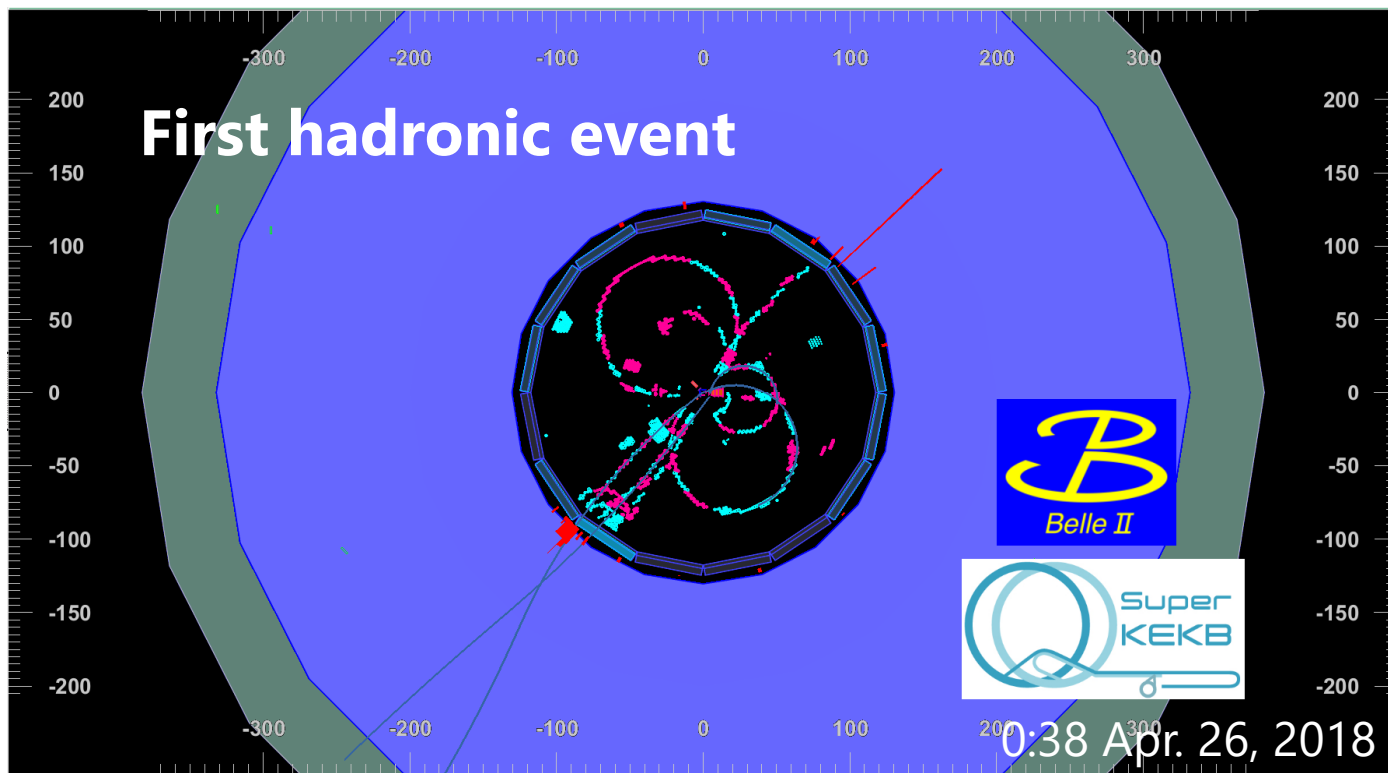


- ~1.5 ppm / 10.5 hrs of current stability obtained.
- Rapid change in AC voltage makes stability worse.

The day before first collision

2018 ...

- Jan. 23: DR LTR beam tuning started.
 - ✓ Major study items for LTR completed at Feb. 7.
- Feb. 8: DR beam tuning started.
- Feb. 10: DR RTL beam tuning started.
- Mar. 16: BT beam tuning started.
- Mar. 19: SuperKEKB HER beam injection and tuning started.
 - ✓ HER captured the beam and RF turn-on on Mar. 20.
- Mar. 27: SuperKEKB LER beam injection and tuning started.
 - ✓ LER captured the beam and RF turn-on on Mar. 30.
- Apr. 20: Collision study with single bunches of e^+ and e^- started.
- Apr. 26: SuperKEKB achieved the collision of e^+ and e^- beams.



Accelerator control room



Belle II control room



Staging in SuperKEKB Commissioning

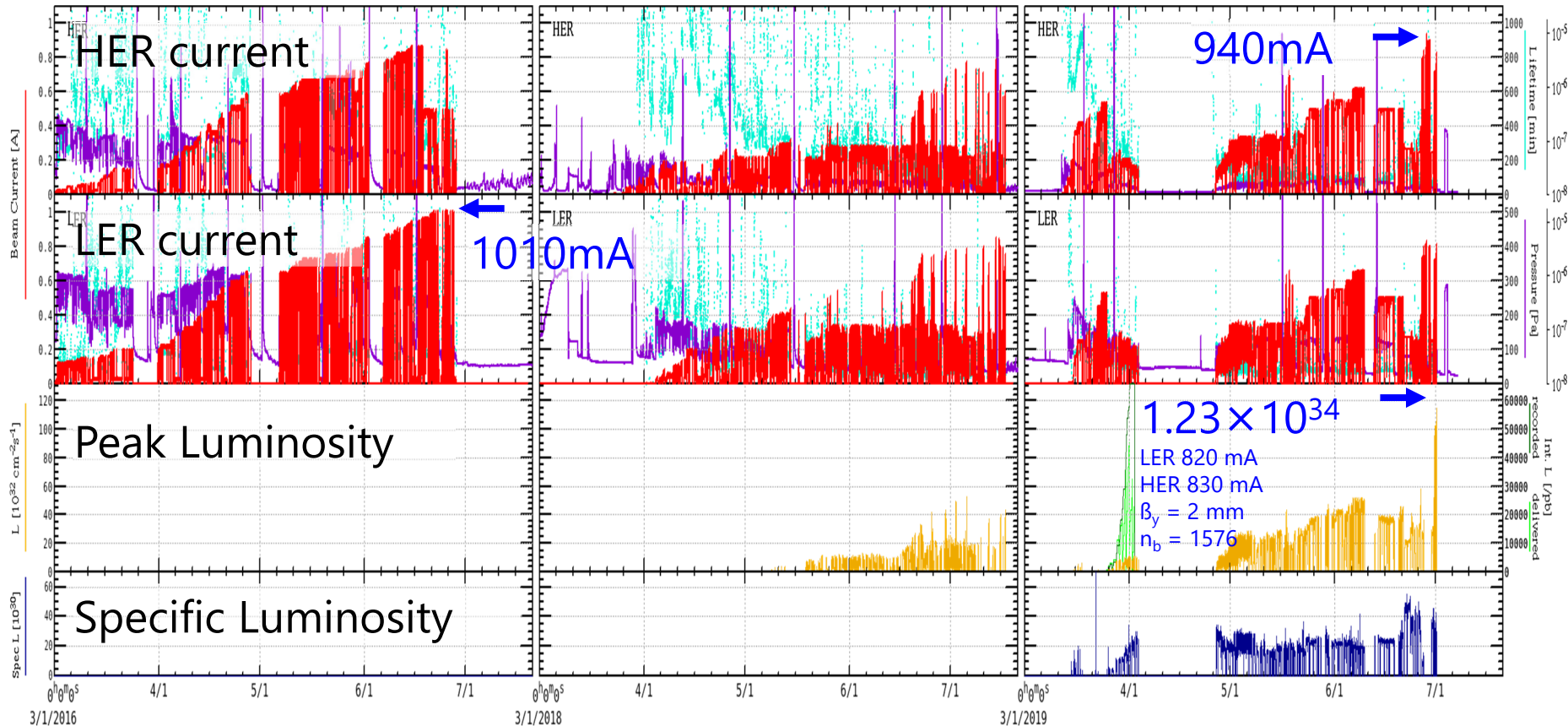
- Phase 1: 2016/02/01 ~ 06/28
Low emittance operation test & vacuum scrubbing without QCS & Belle II detector.
- Phase 2: 2018/03/16 ~ 07/17
“Nano-beam scheme” collision test with QCS & Belle II detector without vertex detector.
- Phase 3-1 (Phase 3; 2019 spring): 2019/03/11 ~ 07/01
Full-scale collider experiment with full Belle II detector.
- Phase 3-2 (Phase 3; 2020 spring): 2019/10/15 ~
High luminosity(peak and integrated) tuning and physics run.

History of Phases 1, 2 and 3 of beam commissioning

Phase 1

Phase 2

Phase 3



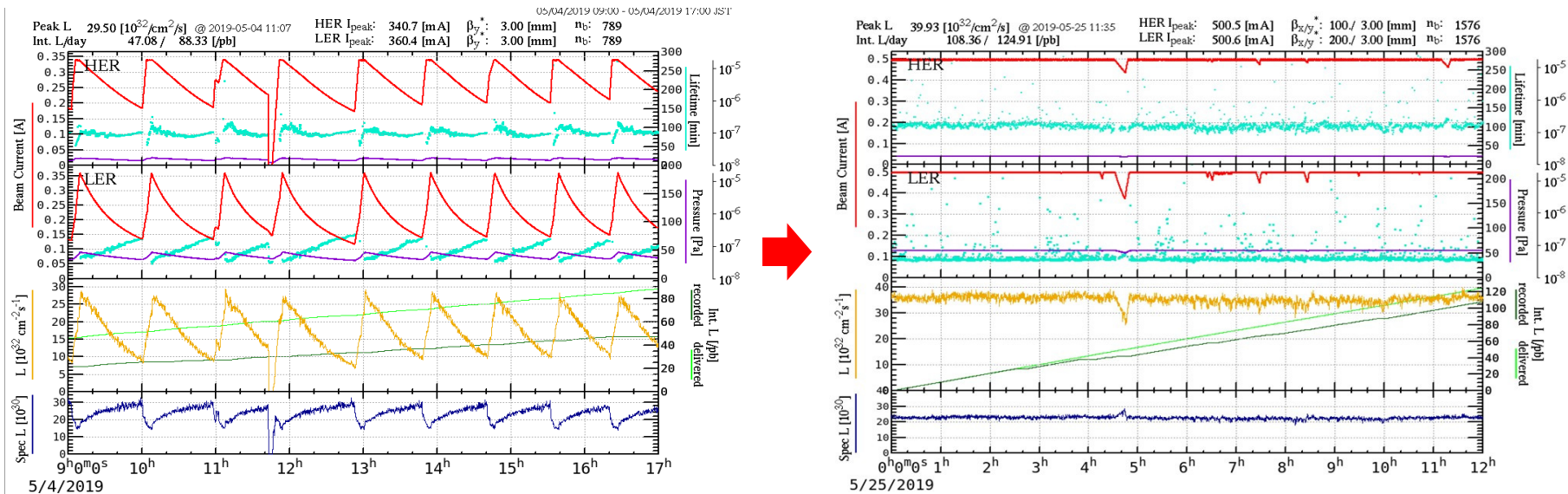
- No collision
- Vacuum scrubbing

- Squeezing
- Collision tuning
- Physics run starts

- Physics run starts with full Belle II
- Further tuning and continuous injection

Continuous injection

In Phase 3, “continuous injection” started, so that e^- and e^+ beams are continuously injected into HER and LER.



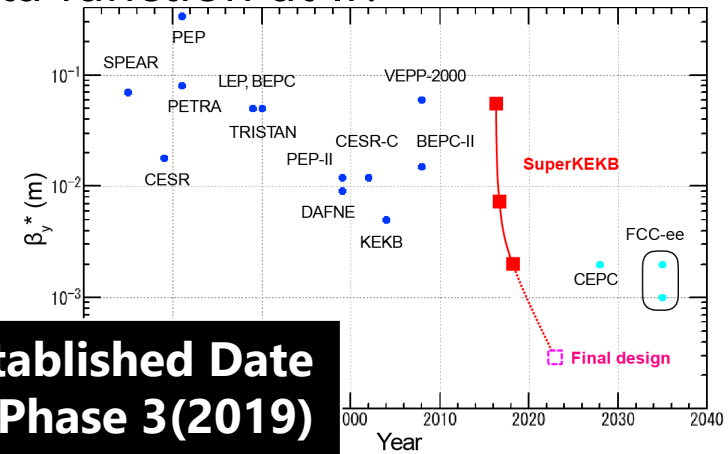
- It was accomplished after successful background reduction by extensive tuning of injection parameters and collimators.
- Stored beam currents are kept constant, and **integrated luminosity is increased.**

Beta squeezing

Luminosity is increased by reducing vertical beta function at IP.

$$L \approx \frac{\gamma_{\pm}}{2e r_e} \left(1 + \frac{\sigma_y^*}{\sigma_x^*} \right) \frac{R_L}{R_y} \xi_{\pm y} \frac{I_{\pm}}{\beta_{\pm y}^*}$$

Beta function is squeezed gradually.



(β_x^*, β_y^*)	Established Date in Phase 2(2018) LER, HER	Established Date in Phase 3(2019) LER, HER
Detuend	3/31, 3/20	3/13, 3/11
(200mm, 8mm)	4/16, 4/10	3/18
(200mm, 6mm)	5/14, 5/10	3/25
(200mm, 4mm)	5/24	3/26
$(200_{\text{LER}}/100_{\text{HER}} \text{mm}, 3\text{mm})$	6/5, 6/20	3/28, 4/1
(80mm, 2mm)	-	6/21

5/14
Continuous Inj.

Detuned → 3mm collision

~ 3months

~ 3weeks

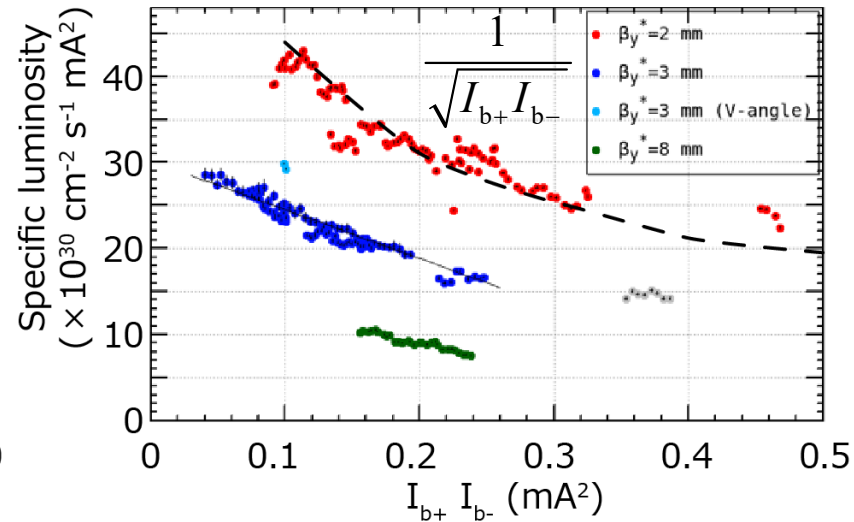
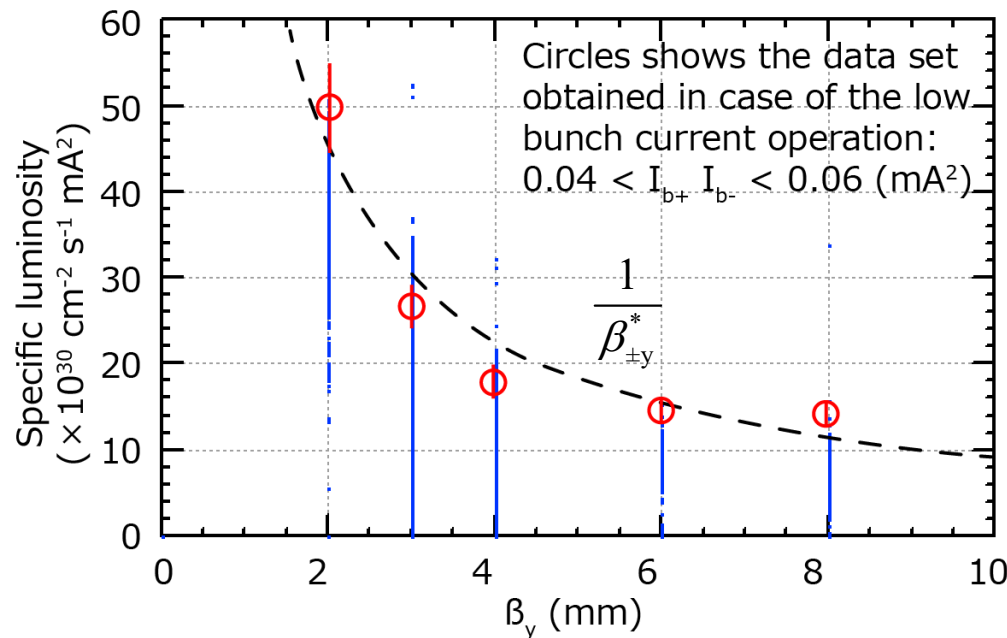
27 QCS quenches
to complete squeezing

No QCS quench

Specific luminosity

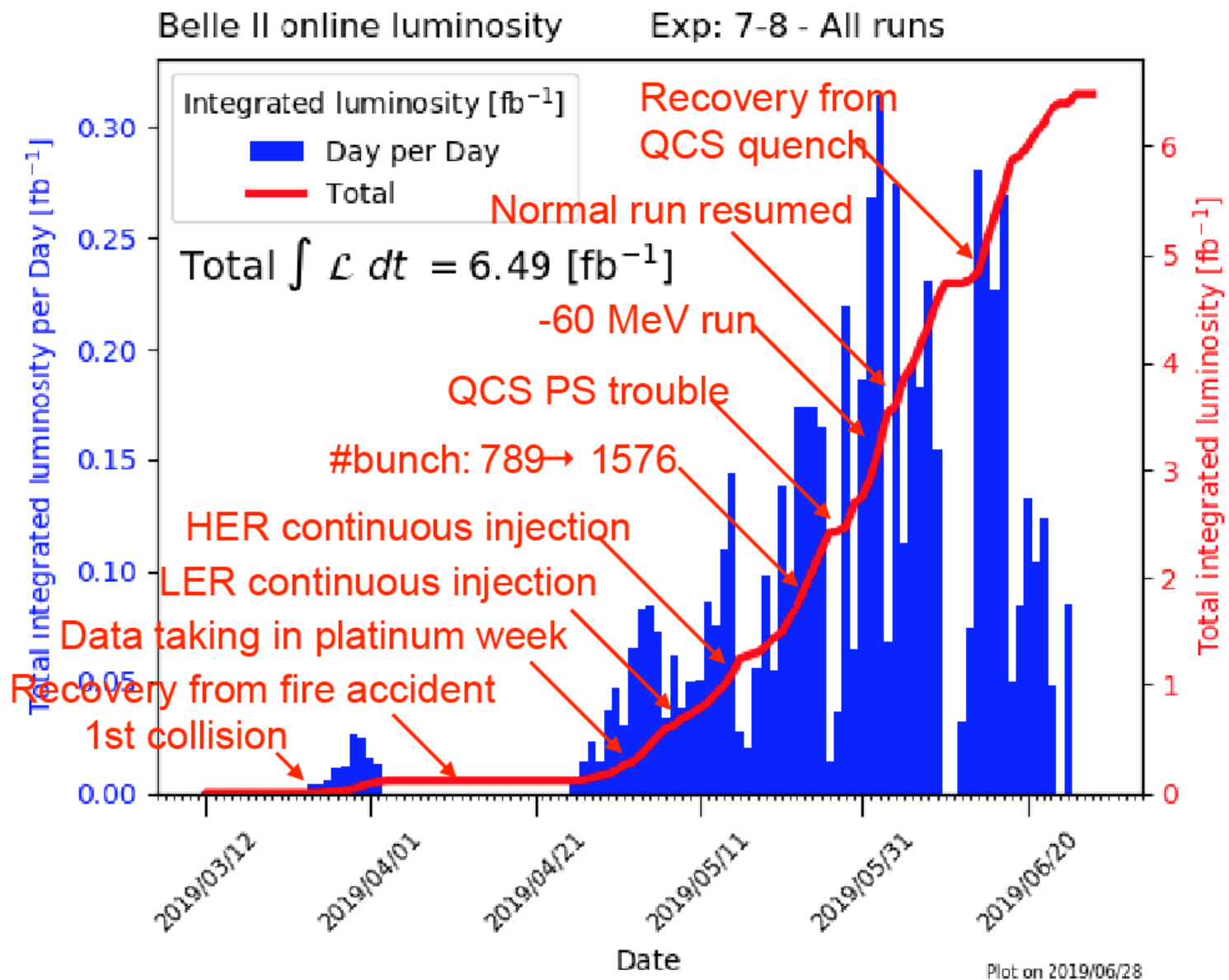
Spec. luminosity: normalized by bunch current product $\times$ # of bunch.

$$L_{\text{sp}} = \frac{L}{I_{b+} I_{b-} n_b} \propto \frac{\xi_{\pm y}}{\beta_{\pm y}^* \sqrt{I_{b+} I_{b-}}}, \quad \text{where } I_{b\pm} = I_{\pm} / n_b$$



L_{sp} is increased by squeezing β_y^* .
($\xi_{\pm y}$ can be kept while squeezing)

Status of physics run in Phase 3 of commissioning



Summary

- SuperKEKB is an electron-positron collider that aims for 40 times higher peak luminosity than that of KEKB.
- After Phase 1 commissioning, QCS and Belle II detector, without vertex detector, were installed in the IR and Phase 2 commissioning with beam collision was started.
- First collision and first hadronic event detected on Apr. 26, 2018.
- After Phase 2 commissioning, vertex detector was installed in Belle II detector and Phase 3 commissioning with full-scale collider experiment was started.
- The “nano-beam scheme” collision was demonstrated down to $\beta_y^* = 2$ mm.
- Continuous injection for both rings is used in regular operation.
- Peak luminosity of $1.23 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ was recorded at $\beta_y^* = 2$ mm, LER 820 mA, HER 830 mA, 1576 bunches.
- Although there are several issues remaining to achieve higher luminosity, we will keep trying to solve them.

Next goal toward summer in 2020

- Increase peak luminosity to $> 3 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ as soon as possible.
 - By squeezing beta by factor of 3.
- The background is low enough to run Belle II detector stably.
 - Tuning of collimators, injection, vacuum scrubbing, etc.